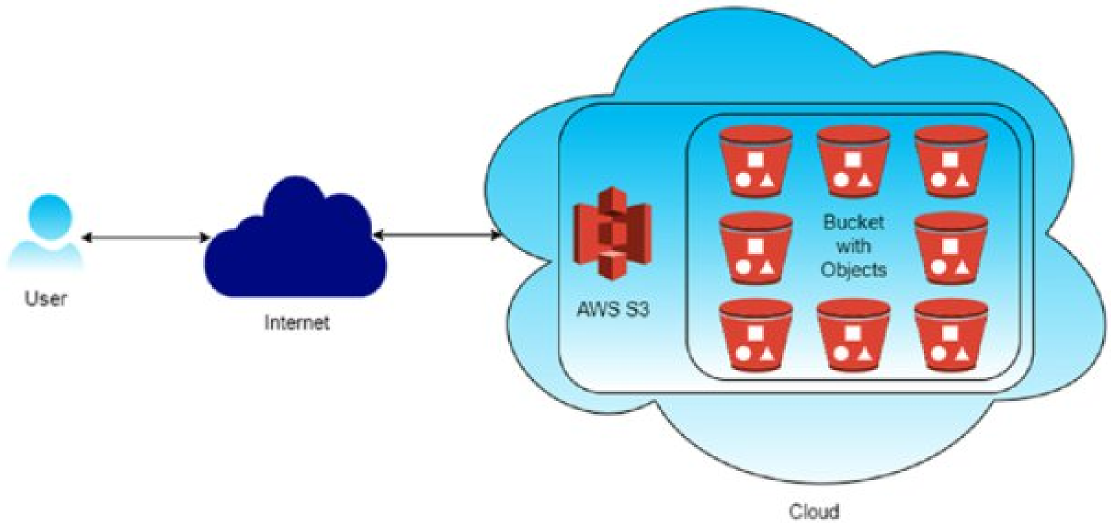
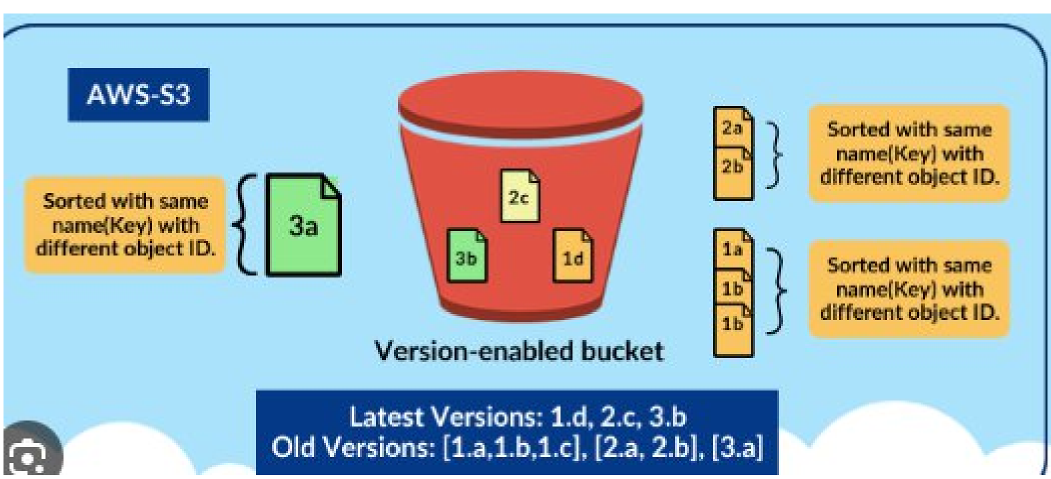
Simple Storage Service (S3)

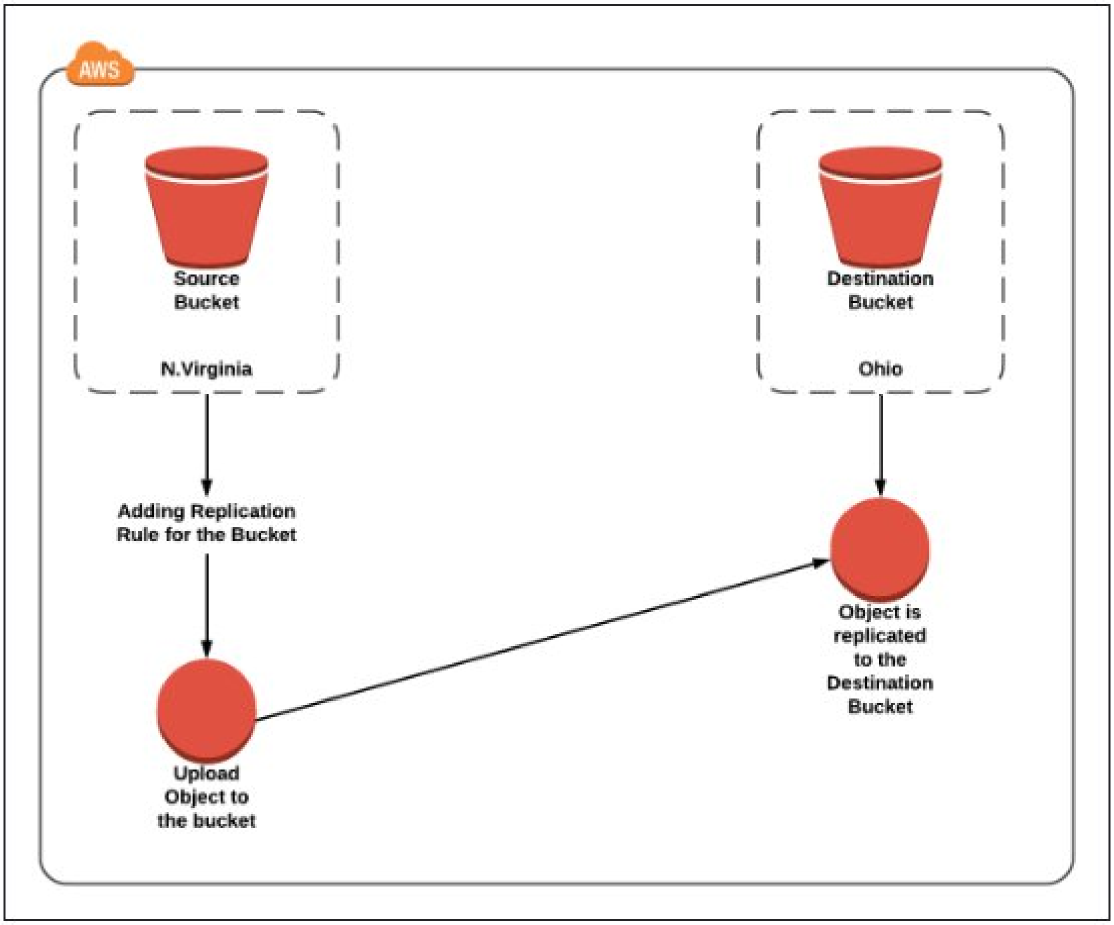


S3 Buckets

1. **Versioning**: Versioning in Amazon S3 is a feature that allows you to keep multiple versions of an object in the same bucket. Whenever you upload a new version of an object with the same key (name) as an existing object, S3 doesn't overwrite the existing object but instead creates a new version of it. This means you can retain and access previous versions of objects, providing an additional layer of protection against accidental deletions or overwrites
   i. **Accidental Deletion Protection** :: With versioning enabled, if you accidentally delete an object, you can still retrieve previous versions of it, preventing data loss.
   ii. **Accidental Overwrite Protection**::Versioning prevents accidental overwrites of objects. Even if you upload a new version of an object with the same name, the previous versions are preserved.



1. **Cross-Region Replication**: Cross-region replication (CRR) in Amazon S3 is a feature that automatically replicates objects from one bucket in one AWS region to another bucket in a different AWS region. This provides redundancy and disaster recovery capabilities, ensuring that your data remains available even if an entire AWS region becomes unavailable.
    i. **Disaster Recovery**: Cross-region replication helps ensure business continuity by replicating critical data to a different geographic region, reducing the risk of data loss due to regional disasters or outages.
    ii. **Low-Latency Access**: Users in different geographic regions can access data from the nearest region, reducing latency and improving performance.
    iii. **Compliance**: Some regulatory requirements mandate data replication across multiple geographic regions for data sovereignty and compliance reasons.
2. **Event Notifications**: S3 can trigger events (e.g., object creation, deletion) that can be routed to AWS Lambda, SNS, or SQS for processing.
3. **Integration**: S3 seamlessly integrates with other AWS services such as AWS Lambda, Amazon CloudFront, AWS Glue, Amazon Athena, and Amazon EMR, enabling you to build powerful data processing and analytics workflows.



1.Creating an S3 bucket, upload an object, and enable public access

1 Create an S3 Bucket (with Public Access Enabled)

Step 1: Open S3 Service

- Login to AWS Management Console
- Go to Services → S3
- Click Create bucket

Step 2: Bucket Configuration

- Bucket name: my-public-bucket-123
- Region: Choose nearest region

Step 3: Disable Block Public Access (IMPORTANT)

Under Block Public Access settings for this bucket:

❌ Uncheck Block all public access

You must uncheck ALL options:

✔ Check I acknowledge that this bucket will become public

☞ Click Create bucket

2️⃣ Enable Public Access at Bucket Level (ACL)

Step 4: Open the Bucket

- Click on the bucket name
- Go to Permissions tab

Step 5: Edit Bucket ACL

- Scroll to Access Control List (ACL)
- Click Edit

Under Public access:

- Everyone (public access) → ✔ Read access

✔ **Save changes**

📌 **This allows public access at bucket level**

3️⃣ Upload Object to the Bucket

Step 6: Upload File

- Go to Objects tab
- Click Upload
- Click Add files
- Select file (e.g., image.jpg)
- Click Upload

4️⃣ Enable Public Access for Object (ACL)

Step 7: Object-Level ACL Permission

- Click on uploaded object
- Go to Permissions tab
- Scroll to Access Control List (ACL)

- Click Edit

Under Public access:

- Everyone (public access) → ✔ Read access

✔ Save changes

5 Verify Public Access

Step 8: Test Object URL

- Copy Object URL
- Open in browser (incognito)

✔ Object should be accessible without login

2. S3 Versioning & Cross-Region Replication

1 Enabling Versioning in S3 (with Output Scenario)

- **Objective**

To enable version control on an S3 bucket and observe object versions.

- **Steps to Enable Versioning**

1. Open AWS Management Console
2. Navigate to S3
3. Select the bucket
4. Go to Properties tab
5. Scroll to Bucket Versioning
6. Click Edit
7. Select Enable
8. Click Save changes

✔ Versioning is now enabled

- **Output Scenario (What You Observe)**

Step A: Upload Object

- Upload file: report.pdf

📌 Output:

- Version ID is assigned automatically

Step B: Modify and Re-upload Same Object

- **Upload report.pdf again (same name)**

📌 Output:

- Old version is preserved
- New version becomes current

Step C: Delete Object

- Click Delete on report.pdf

📌 **Output:**

- Object appears deleted
- A Delete Marker is created
- Older versions still exist and can be restored

2️⃣ Cross-Region Replication (CRR)

🔹 **Objective**

To replicate objects from one bucket to another in a different region using AWS-managed permissions.

🔹 **Prerequisites (Very Important)**

✔ Versioning must be enabled on both buckets

✔ Source and destination buckets must be in different regions

✔ AWS Academy automatically handles replication permissions

Step 1️⃣ Create Destination Bucket

1. Go to S3 → Create bucket
2. Bucket name: dest-crr-bucket
3. Select different region
4. Complete bucket creation
5. Enable Versioning (same steps as above)

Step 2 Create Replication Rule (Without IAM Selection)

1. Open Source bucket
2. Go to Management tab
3. Click Replication rules
4. Click Create replication rule

Step 3 Configure Replication Rule

- Rule name: academy-crr-rule
- Status: Enabled
- Scope: Apply to all objects

Step 4 Choose Destination Bucket

- Select Another bucket
- Choose Destination bucket
- Confirm different region

📌 **AWS Academy Note:**

You will see an option like:

"AWS will create and manage the required permissions automatically"

✔ Select Use AWS-managed role / default role

Step 5 Replication Options

- **Replicate:**
  - ✔ New objects
  - ✕ Existing objects (optional, depends on Academy permissions)
- **Encryption: Leave default**

Click Save

✔ Replication rule is now active

3 CRR Output Scenario (What You Observe)

❖ Step A: Upload New Object to Source Bucket

- Upload file: image.png

📌 Output:

- Object is uploaded normally

◆ **Step B: Verify Destination Bucket**

- Open destination bucket
- Object image.png appears automatically
- Version ID is present

✔ **Replication successful**

3. Static Website Hosting in S3

(Using Bucket-Level & Object-Level Permissions – ACL Method)

◆ **Objective**

To host a static website in Amazon S3 using index.html and error.html, and allow public access using ACL-based bucket and object permissions.

◆ **Prerequisites**

- AWS / AWS Academy account
- Two files:
  - index.html
  - error.html

1 Create an S3 Bucket

1. Login to AWS Management Console
2. Go to Services → S3
3. Click Create bucket
4. **Enter:**
   - Bucket name: my-static-site-bucket-123
   - Region: Nearest region
5. **Under Block Public Access:**
   - ✕ Uncheck Block all public access

- ✔ Acknowledge the warning

6. **Click Create bucket**

✔ Bucket created successfully

2 Upload Website Files

1. Open the bucket
2. Go to Objects tab
3. Click Upload
4. Click Add files
5. Select:
   - index.html
   - error.html
6. Click Upload

✔ Files uploaded

3 Enable Static Website Hosting

1. Go to Properties tab
2. Scroll to Static website hosting
3. Click Edit
4. Select Enable
5. Choose Host a static website
6. Enter:
   - Index document: index.html
   - Error document: error.html
7. **Click Save changes**

✔ Static website hosting enabled

4 Enable Bucket-Level Public Access (ACL)

- **Purpose**

Allows public users to access objects inside the bucket.

1. Go to Permissions tab

2. Scroll to Access Control List (ACL)
3. Click Edit
4. Under Public access:
    - Everyone (public access) → ✔ Read
5. Click Save changes

✔ Bucket-level public read access enabled

5 Enable Object-Level Public Access (ACL)

Steps (For Each File)

1. Go to Objects tab
2. Click on index.html
3. Go to Permissions
4. Scroll to Access Control List (ACL)
5. Click Edit
6. Under Public access:
    - Everyone (public access) → ✔ Read
7. Save changes

🔁 **Repeat the same steps for error.html**

✔ **Object-level public access enabled**

6 Access the Static Website

1. Go to Properties
2. Scroll to Static website hosting
3. Copy the Bucket website endpoint
4. http://my-static-site-bucket-123.s3-website-region.amazonaws.com
5. Paste into browser

✔ index.html page loads successfully

7 Error Page Output Scenario

- ◆ Test Error Page
- Open an invalid URL:
- http://bucket-name.s3-website-region.amazonaws.com/invalid.html

📌 Output:

- error.html page is displayed

✔ Error page working correctly

Amazon S3 – Scenario-Based Questions

1. A developer uploads an object to an S3 bucket but cannot access it from the browser, even though the bucket exists. What permission-related issue could be the cause?
2. An S3 bucket policy allows public read access, but a specific object inside the bucket is still not accessible. Why might this happen?
3. A bucket has **no bucket policy**, but individual objects have public ACLs. Will users be able to access those objects? What determines the final access decision?
4. A company wants to ensure **no public access** to any object in an S3 bucket, even if someone mistakenly applies a public ACL. What S3 feature should be enabled?
5. You attempt to create an S3 bucket named my-company-data but receive a naming error. What S3 bucket naming rules might have been violated?
6. Two different AWS accounts try to create a bucket named project-backup-2026. One succeeds, the other fails. Why?
7. You enabled **static website hosting** on an S3 bucket and uploaded files, but accessing the website URL returns **403 Forbidden**. What is the most likely cause?
8. Static website hosting is enabled, but accessing the root URL shows an error saying **index.html not found**. What is the actual issue?
9. What HTTP error is returned when the **index document is missing** in S3 static website hosting, and why does this happen?
10. A bucket policy allows public access, but **static website hosting still doesn't work**. What object-level setting must also be verified?
11. You accidentally overwrite an important object in S3. Versioning was enabled earlier. How can the original file be recovered?
12. A user deletes an object from a versioned S3 bucket. Is the data permanently removed? Explain what actually happens.

13. You enabled Cross-Region Replication (CRR) on a bucket, but objects are not replicating. What configuration might be missing?
14. After enabling CRR, only newly uploaded objects are replicated, but old objects are not. Why?
15. A company wants to replicate S3 data to another region for **disaster recovery and compliance**. What are the key prerequisites for enabling CRR?